

Django concepts — as actually used in this project

Models = tables

A Django model class is a PostgreSQL table; each instance is a row. `TimeStampedModel` (core app, abstract) gives every child `created_at` / `updated_at` columns — abstract matlab: columns COPY ho jaate hain child ki table mein, koi extra JOIN nahi.

ForeignKey + on_delete — the deletion contract

- **PROTECT** (our default for money/history anchors): parent delete = ERROR. Ledger row ke peeche ka worker/Adda/SWA kabhi gayab nahi ho sakta.
- **CASCADE**: child is meaningless without parent (e.g. LayeringRollEntry dies with its stage record — it was per-stage scratch).
- **SET_NULL**: optional reference (e.g. evidence link) — parent ja sakta hai, child reh jata hai. Rule of thumb yahan: paise ya history touch karta hai? → PROTECT.

Constraints — DB as the last guard

`UniqueConstraint(condition=...)` (partial unique): ≤ 1 ACTIVE task per (stage_record, worker) while cancelled history accumulates. `CheckConstraint`: `finalized  $\Rightarrow$  settled_at`, `recovered  $\leq$  outstanding`, XOR exactly-one-parent. App bug ho bhi jaye, DB jhooth store nahi karega. (Django 5.0.1 note: partial unique uses `condition=`, checks use `check=`.)

Managers

`.objects` = default; `.active` (ActiveManager) = opt-in soft-delete filter. Default ko override NAHI kiya — warna rows chupke se gayab dikhte.

Frozen snapshot vs live SUM (the project's signature pattern)

Snapshot column = “us waqt kya tha” (`expected_rate`, `processing_cost`, `AddaSettlementItem`) — write-once, kabhi recompute nahi. Balance = “abhi kya hai” — HAMESHA ledger SUM. Dono mix karoge to ya audit tootega ya accuracy.

Migrations

Schema changes = migration files (append-only history of the schema itself). Data migrations backfill ONLY known facts (owner rule) — states kabhi invent nahi karte. Destructive ones get clone rehearsal first.

No signals (ADR-0001)

`inventory/signals.py` is a tombstone. Everything happens explicitly inside service functions — “yeh kab chala?” ka jawab hamesha stack trace deta hai.